

Optimal Switch and Tie Line Planning in Distribution Networks: Benchmarking a Practical MILP Model With a Fast Heuristic Approach

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Abstract—Optimal placement of switches and tie lines is an integral task in distribution system planning. Owing to the problem complexity and the presence of binary decision variables, using heuristic methods for obtaining a close to optimal switch and tie line plan is a common practice in industry. Efforts to employ mixed-integer linear programming (MILP) to guarantee the solution optimality also tend to sacrifice the modeling accuracy through oversimplifying assumptions to make the problem tractable. Aiming to tackle these challenges, we present an efficient, yet accurate, MILP model for optimal switch and tie line planning. The proposed model avoids common simplifying assumptions in the state-of-the-art MILP models while preserving the solving efficiency. In order to demonstrate the applicability and scalability of the proposed MILP approach, it is applied to multiple test networks, and the results are compared with those of a fast heuristic model. The outcomes not only represent the high efficiency and accuracy of the MILP model but also validate the close to optimality of the heuristic approach developed for practical applications as a module in a commercial distribution system planning toolbox.

Index Terms—Electricity distribution system, mixed-integer linear programming, optimization, switch planning, tie line.

NOMENCLATURE

Indices

a Index for restoration schemes.

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e	Index to determine sending and receiving ends of line sections.
k	Index signifying the type of switches.
$l, \bar{l}$	Indices for line section.
n	Index for load nodes.
$r, \bar{r}$	Indices for potential tie lines.
Sets	
$A_{l,n}, A_{r,n}$	Set of all possible ways for restoring the power of load node n during the outage of line section l and tie line r , respectively.
E	Index set $\{es, er\}$, where es and er signify the sending and receiving ends of a line section.
E_a	Subset of E indicating the location of the switch in the faulted line section that can be leveraged for fault isolation in restoration plan a .
$\bar{E}_a$	Subset of E determining which side of a tie line must be normally closed to require restoration scheme a after the failure in the tie line.
K	Index set $\{F, M, R\}$, where F , M , and R respectively stand for field circuit breakers, manual switches, and remote-controlled switches.
L	Set of all line sections.
L_a	Set of line sections that can be switched off to isolate the faulty line section in restoration scheme a .
N	Set of all load nodes.
R	Set of all tie lines that are candidates for construction.
R_a	Set of tie lines that can be used in restoration scheme a .
Parameters	
BM	Sufficiently large number.
D_n	Power demand at load node n .
$ER_{l,n}, ER_{r,n}$	Binary parameters, which are equal to 1 if node n is the receiving end of respectively line section l and tie line r , being 0 otherwise.
$ES_{l,n}, ES_{r,n}$	Binary parameters, which are equal to 1 if node n is the sending end of respectively line section l and tie line r , being 0 otherwise.
g	Annual rate of demand growth.
IO^k	Total investment and lifetime operating cost of a switch of type k .

$IO^{St,k}$	Total investment and lifetime operating cost of a switching station of type k .
IO_r^{Tl}	Total investment and lifetime operating cost for candidate tie line r .
T	Planning horizon.
α	Annual interest rate.
γ_n^E, γ_n^F	Customer interruption costs respectively per unit of curtailed energy and per unit of curtailed power per interruption at load node n .
δ	Present value factor for customer interruption costs.
η_n	Number of customers at load node n .
λ_l, λ_r	Failure rate of line section l and tie line r , respectively.
$\xi_{l,n}$	Binary parameter which is equal to 1 if line section l and node n are in the same feeder, being 0 otherwise.
π_l, π_r	Repair times for line section l and tie line r .
π^M, π^R	Manual and remote switching times.
Variables	
$c_n^{St,k}$	Investment and lifetime operating cost of switching station of type k at load node n .
CICE, CICF	Energy and frequency oriented customer interruption costs, respectively.
IO^{St}, IO^{Sw}	Total investment and operating costs of respectively switching stations and switches over their useful lifetime.
$x_{l,e}^k, x_{r,e}^k$	Binary investment variables for installing a switch of type k at location e of line section l and tie line r , respectively.
x_r^{Tl}	Binary investment variable for candidate tie line r .
$y_{r,e}^{NO}$	Binary variables representing the normally-open side of the tie lines.
$\theta_{l,n,a}, \theta_{r,n,a}$	Annual power interruption duration for load node n due to the failures in line section l and tie line r , respectively, if adopting restoration scheme a .
$\nu_{l,n}, \nu_{r,n}$	Annual frequency of power interruptions for load node n due to the failures in line section l and tie line r , respectively.
$\tau_{l,n}, \tau_{r,n}$	Annual power interruption duration for load node n due to the failures in line section l and tie line r , respectively.
$\phi_{l,n,a}, \phi_{r,n,a}$	Annual frequency of power interruptions for load node n due to the failures in line section l and tie line r , respectively, if adopting restoration scheme a .
$\chi_{l,n,a}^\nu, \chi_{r,n,a}^\nu$	Auxiliary binary variables utilized to determine the best way to isolate faulty line section l .
$\chi_{l,n,a}^\tau, \chi_{r,n,a}^\tau$	Auxiliary binary variables utilized to determine the best way to restore load node n after failure in line section l .

I. INTRODUCTION

OPTIMAL switch placement methods in distribution networks take into account a multitude of driving factors, often grouped into investment costs, running costs, and interruption costs. Overly emphasizing one affects the others, sometimes adversely and sometimes beneficially. A central attribute of such a planning method must then be to consider these major factors in a balanced fashion, whilst also complying with the core technical and operational constraints of a distribution network. In a wider network planning context, reliability, including switch placement, should be integrated in the planning process, as mitigating interruption costs has a significant impact on the topology of a distribution network. This consideration has motivated the heuristic algorithm presented in this paper, which has been devised specifically to be computationally fast, as it is executed potentially millions of times in a distribution network planning algorithm. It has been claimed in earlier publications, e.g., in [1], that this heuristic methodology is close to optimum. This paper will challenge that claim by comparing the relatively crude though comprehensive heuristic algorithm with a state-of-the-art mixed-integer linear programming (MILP) model.

Owing to the importance of reliable power distribution, over the past decades, electrical engineers and power system researchers have tried to find the most efficient sectionalizing switch (SS) placement in distribution networks through various methods. Traditionally, the distribution utilities tend to use their engineers' experience or rule-of-thumb methods to determine the optimal placement of SSs in their networks. This has been mainly due to the limited computational capability of processors and limitations of mathematical modelling. Note that optimal switch placement problem has a combinatorial nature and is classified as a non-deterministic polynomial-time (NP)-hard problem [2]. In the 1990s and 2000s, however, many research studies were published on the mathematical modelling of this planning problem, employing various heuristic and metaheuristic algorithms to solve it. Some examples are [3], [4], [5], [6], which used a genetic algorithm, ant colony optimization, artificial immune system, and graph-based solution, respectively, to solve the problem. However, in the 2010s, immense growth in the computational capacity of processors together with significant advancements in MILP solvers led to developing MILP models for the optimal switch allocation in distribution networks [7]. Of course, developing a tractable MILP formulation meant that some aspects of this problem, e.g., candidate locations and types of switches, lateral sections, multiple tie lines per feeder, concurrent optimization of tie lines, and failures in tie lines, should be simplified. Yet, such models can be easily formulated for various cases and solved using off-the-shelf solvers in a finite amount of time, while the convergence to global optimum can be guaranteed. Contrary to these efforts, authors in [8], [9] employed novel heuristic algorithms to solve more accurate models for the switch placement problem in distribution networks, while considering practical aspects of the problem simplified in the MILP models. However, such heuristic methods cannot guarantee the convergence to the global optimal solution;

nor can they guarantee the found local optimum is *close enough* to the best possible solution.

In the seminal work of Abiri-Jahromi et al. [7], one of the first MILP-based models was developed for the reliability-oriented optimal switch placement problem that could guarantee reaching the global optimal solution. In their model, the optimization minimizes the aggregated costs of remote-controlled switch (RCS) installation and customer outages. Despite the numerous benefits of such an easy-to-solve model, it included many simplifications that made it less applicable in practice. Accordingly, numerous research studies in the past few years have tried to extend the previous models or develop new ones with MILP formulations, while taking into account important practical aspects of optimal SS placement. Considering earth faults in networks, Siirto et al. in [10] extended the MILP model originally proposed in [7]. Heidari et al. in [11] developed an optimal SS placement model while taking into account the distributed generation in the system. In [12], the stochastic nature of faults in the system was modelled in an MILP-based SS placement model. Addressing two drawbacks of previous studies, i.e., RCS as the sole installation option and considering switch malfunction probability, Farajollahi et al. in [13] developed a MILP model for the optimal placement of both the manual switches (MSs) and RCSs. They also proposed an MILP formulation in [14] for the simultaneous allocation of fault indicators, MSs, and RCSs. Another simplified aspect of MILP optimal SS placement models was disregarding the necessary investments in tie lines and their SSs for an efficient reliability-oriented network planning. Studies [15], [16] focused on addressing this issue, where MILP models were developed to find the optimal solution for the installations of tie lines and SSs simultaneously. Still, one major issue in all these MILP studies has been their inability to consider installation of switches in the lateral sections of main distribution feeders, as well as the possibility of faults in those sections. Recent studies, such as [17], [18], [19], addressed these drawbacks and proposed efficient MILP models that could consider the complex topology of distribution networks. While the optimization models in [17], [18] cannot consider installation of new tie lines in the networks as an option, the ones in [19], [20] are able to do so. Additionally, the model presented in [18] relies on an approach based on “fictitious power flow”, which requires formulating post-fault network reconfiguration for every contingency scenario. This modeling concept, albeit being powerful and flexible, can result in the rapid growth of the problem dimension with respect to the network size. This, in turn, can cause intractability issues in the case of large networks, as reported in [21] for a distribution expansion planning setting.

One can see the significant improvements in the accuracy and thoroughness of MILP models for the optimal placement of protective devices in distribution networks in recent years, which has been achieved thanks to the incremental advancements of multiple research studies. This has made them roughly comparable with the heuristic methods and algorithms that are being used in industry by distribution utilities and other related companies.

Nevertheless, to the best of our knowledge, none of the existing MILP models for the optimal placement of protective devices

in distribution networks [7], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [22] consider all the practical aspects of the problem that a typical industry’s heuristic algorithm can efficiently model. This is mainly because formulating those aspects in a MILP fashion is either too difficult or too inefficient making the MILP problem intractable. One main aspect that must exist in a practical switch placement model is considering the possibility of faults in the laterals of main distribution feeders and allowing installation of switches in them. The importance of this issue has been already demonstrated in [23]. However, the MILP models proposed in [7], [10], [11], [12], [13], [14], [15], [16], [22] disregarded this aspect, only considering main feeder sections of the distribution network. The second necessary feature of a SS placement model to make it applicable in real-world applications is considering the installation of new tie lines and their respective switches in the optimization problem. This should be done together with the placement of switches as the placement of each group would impact the optimal allocation of the other. Therefore, to reach the optimal solution, the investment optimization of both switches and tie lines should be simultaneously considered [15]. The models in [7], [10], [11], [12], [13], [14], [17], [18], [22] do not consider this aspect. The third important feature is to have all types of switches, which are typically used for isolating faults in distribution networks, as the available options for installation, including MSs, RCSs, and field circuit breakers (FCBs). Similar to the previous feature, this is because the optimal allocation of each switch type is dependent on the placement of the others, so the optimization of all should be done at the same time to achieve the optimal switch placement [18], [24]. This feature is not modelled in [7], [10], [11], [12], [13], [14], [15], [16], [17], [19], [20], [22]. The fourth important feature is taking into consideration the potential failures in newly built tie lines, as well as the practical complications of installing switches on them. In practice, to monitor the tie switch availability for managing faults, the tie line should be energized. This is done by one end of tie line being connected to a feeder and the other end open to respect the distribution network radiality. Consequently, in the case of a fault in the tie line, the feeder connected to the normally-closed side will be impacted as well. The models in [15], [19], [20], while considering tie line installation, ignored such practical matters. This can lead to over-optimistic optimal solutions for real-world SS placement problems, mainly because of not considering the faults in tie lines and their effect on customer interruptions [16]. In addition, those models cannot help the planner with deciding on which end of the tie lines the switches should be installed or whether one or two switches on an installed tie line is the optimal choice. Lastly, none of the previous studies have considered the installation of switching stations [7], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [22]. Equipping a feeder section with an RCS or an FCB requires installing a switching station that includes the equipment needed for the operation of RCSs and FCBs. In this context, simply including the cost of a switching station in the switch investment cost is incorrect, since a single switching station at a node can support any number of switches on the feeder sections or tie lines to which the node is connected.

TABLE I
COMPARISON OF MILP-BASED SWITCH OPTIMIZATION MODELS

Feature	[12]–[15], [22]	[16]	[18]	[17], [19], [20]	[24]	Proposed model
I	X	X	✓	X	✓	✓
II	X	X	X	X	X	✓
III	X	X	✓	✓	X	✓
IV	X	✓	X	X	X	✓
V	X	X	X	X	X	✓

I: Considering FCB as a switch option

II: Tie line installation considering practical aspects

III: Modeling feeder laterals for complex network topologies

IV: Modeling failure of tie lines

V: Modeling switching stations

In the recent literature, there are notable studies, such as [21], [25], [26], [27], that consider or assess the reliability of distribution networks using MILP in the planning stage. These studies, however, do not solve the optimal switch placement problem. They assume that all line sections are already equipped with manual switches, and a circuit breaker is installed at the sending end of each feeder. These assumptions are made to keep the MILP models tractable since their main focus is on the expansion planning of distribution network, rather than optimal switch placement [21], [25], [26], [27].

This paper proposes a novel MILP model for the optimal SS placement problem, where all the aforementioned aspects are considered together in an efficient manner. Building upon the fundamental concepts introduced in [20], the modeling framework proposed in this paper leverages a data transformation stage to characterize all possible restoration schemes. This facilitates mathematical modeling when dealing with practical planning and operation aspects such as complex configurations, tie line installation, as well as modeling switching stations, tie line failures, and operational details regarding fault mitigation and power restoration. The proposed model can determine the allocation of SSs, FCBs, and tie lines in distribution networks with complex topologies (i.e., both main feeders and their laterals can be considered) while accounting for the investments in switching stations required for operating RCSs and FCBs. This model does not make any simplifying assumption regarding the candidate locations for installation of SSs, FCBs, or tie lines; available protective device options; and the type of sections in which a fault might happen (e.g., main feeder, lateral, or tie line). Moreover, due to the MILP formulation of the model and its efficiency, the optimal switch placement problem can be easily formulated and solved by commercial solvers in a finite amount of time, guaranteeing convergence to the global optimum.

Table I summarizes the main differences between the MILP model proposed in this paper and those presented in the closely relevant literature. In this table, symbols “✓” and “X” respectively indicate whether or not a particular feature is considered.

To demonstrate the effectiveness of the proposed MILP model in practical applications, we compare a quick and heuristically constrained benefit-to-cost switch optimization algorithm built specifically for a distribution network planning tool with our state-of-the-art MILP model that achieves the theoretical optimum for the placement and optimization of line switches. The question is, “how much better is the global optimum over the close-to-optimum algorithm used in the planning algorithm,

and what is the computational burden imposed?” Both algorithms can handle three switch types, load breaking manual and remote-operated switches, and network circuit breakers. The costing of customer interruptions used in each method, itself a major research topic, has been harmonized to enable an accurate comparison. The networks considered are highly realistic. Whilst confidentiality prohibits the use of actual network data, the planning algorithm [28] (that employs the cruder switch optimization algorithm) has been used to produce three solution networks that use synthetic but realistic data, a mixed (urban, suburban, and rural), an urban, and a rural, respectively referred to as 1000-node, 230-node, and 186-node test networks in this paper.

II. PROPOSED MILP MODEL FOR OPTIMAL SWITCH AND TIE LINE PLANNING

This section is devoted to the proposed MILP model for optimizing the planning of switches and tie lines. The model is applicable to distribution networks with multiple tie lines and lateral sections per feeder. Aside from the sending end of the outgoing main feeders, which are equipped with circuit breakers, it is assumed that both ends of all line sections are candidate locations for installing switches. Also, a set of candidate branches for adding tie lines is considered. The aim is to find the optimal decisions concerning the type of switch, if any, to be installed at every candidate location, and the optimal locations for constructing tie lines. The model also finds the optimal types of switches at each end of the installed tie lines and their normally-open side. The types of switches considered in this paper comprise FCB, RCS, and MS. In contrast to MSs and RCSs, FCBs, which are more expensive, can interrupt fault current almost instantaneously, thereby reduce both the frequency and duration of power interruptions [24]. Featuring a shorter switching time as compared with MSs, RCSs can substantially decrease duration of power outages, yet their investment and operation costs are higher. Thus, the proposed model optimizes the types of switches and their locations to balance the reliability and investment and operation costs of switches.

As a common practice in the relevant literature [7], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [25], [26], [27], we only consider sustained interruptions of line sections. Such interruptions are typically characterized by two stochastic parameters, namely failure rate and repair time, whose expected values are used in the analytical distribution system reliability assessment [7], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [25], [26], [27]. An illustrative example of a tiny distribution system with laterals and multiple candidate tie line locations is depicted in Fig. 1. A restoration scheme (indexed by a) for faulted line section l and load node n is a combination of switching operations to restore power at node n when a failure occurs in line section l . Each restoration scheme is characterized by three sets, namely E_a , L_a , and R_a (in the case of tie line failures, $\bar{E}_a$ is also needed). As there can be multiple schemes for restoring node n when line section l fails, $A_{l,n}$ is defined as the set of all restoration schemes associated with l and n .

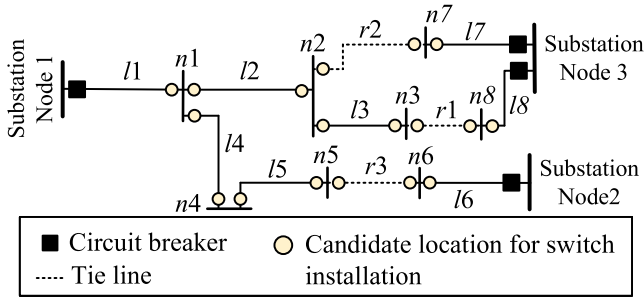


Fig. 1. An illustrative distribution system.

Taking an example from the tiny system depicted in Fig. 1, for line section $l1$ and load node $n1$, there is one possible restoration scheme to restore power to the load node if the line section fails. It involves isolating the faulted section $l1$ using its receiving-end switch and subsequently rerouting the power to $n1$ via one of the tie lines $r1$ – $r3$ to preserve the network radiality. This specific scheme is defined by $E_a = \{er\}$, $L_a = \emptyset$, and $R_a = \{r1, r2, r3\}$, meaning that no other line switches are utilized for isolation, apart from the receiving-end switch of $l1$. The complexity of restoration schemes can vary significantly across different line and node combinations. For example, in the case of faulted section $l1$ and load node $n3$, there are three discernible restoration schemes: 1) isolating by the receiving-end switch of $l1$ and restoring the power via tie lines $r1$ to $r3$, characterized by $E_a = \{er\}$, $L_a = \emptyset$, and $R_a = \{r1, r2, r3\}$, 2) employing a switch on line $l2$ and restoring the power via tie lines $r1$ or $r2$, characterized by $E_a = \emptyset$, $L_a = \{l2\}$, and $R_a = \{r1, r2\}$, and 3) using a switch on line $l3$ to isolate the fault and restoring the power via tie line $r1$, characterized by $E_a = \emptyset$, $L_a = \{l3\}$, and $R_a = \{r1\}$. Restoration schemes for tie lines depend also on their operating condition, i.e., the normally-open sides of the tie lines that are unknown a priori as they are optimized in the model. Thus, for tie lines, apart from E_a , L_a , and R_a , set $\bar{E}_a$ is needed to characterize restoration scheme a . As an example, let us consider the restoration of $n6$ following a failure in $r3$ in Fig. 1. Assuming that $n5$ and $n6$ are arbitrarily selected as the sending and receiving ends of $r3$, respectively, there is only one restoration scheme characterized by $E_a = \{er\}$, $L_a = \emptyset$, and $R_a = \emptyset$, an $\bar{E}_a = \{er\}$. The restoration scheme implies that if the receiving end of $r3$ (the side connected to $n6$) is normally closed, failures in $r3$ can affect $n6$, and to mitigate that a switch on the receiving end of $r3$ is needed. Note that depending on the switch type, the impact of failure on $n6$ varies, e.g., if an MS is installed at the receiving end of $r3$, $n6$ is restored after a manual switching operation to isolate $r3$.

Thus, these schemes highlight various potential paths for restoration, depending on the switches and tie lines available. Note that, through restoration schemes, we only determine restoration “possibilities”. The actual restoration time depends on the types of installed switches in each restoration scheme. Where multiple schemes exist for restoring a load node following a feeder section failure, the best scheme should be chosen. These are considered in the mathematical model presented in the following sections. The definition of restoration schemes along

with the operation conditions enforced by the mathematical model guarantee network radiality after post-fault reconfiguration. However, the presented model disregards the power flow constraints, which is acceptable for distribution switch planning purposes [7], [10], [11], [12], [13], [14], [15], [16], [20], [22], [23], [24]. Nonetheless, such limitations can be accounted for by applying power flow to post-fault network configurations in various restoration schemes and excluding those that violate the limits. Also, the distribution system planning method that was used to produce the realistic networks utilized in this paper does consider all the technical constraints.

Lastly, it is worth noting that the number of potential restoration schemes has a linear relationship with the size of the network. This relationship is influenced by the quantity of line sections within a feeder, as well as the number of available tie lines. The restoration possibilities for a feeder are insulated from other feeders within the network and exclusively depend on its internal configuration of line sections, nodes, and tie lines. Consequently, if the network’s size, quantified by the count of feeders, increases by a factor of n , the total number of restoration schemes across the network also increases proportionately by the same factor.

A. Mathematical Formulation of the Optimization

This subsection presents the mathematical formulation of the proposed optimization model for switch and tie line placement in a distribution network. In this optimization problem, we aim to minimize the objective function, (1), comprising the total investment and operation costs of switches, tie lines, and switching stations, as well as the present value of customer interruption costs, consisting of curtailed-energy-oriented cost $CICE$ and interruption-frequency-oriented cost $CICF$, over the planning horizon. The interruption costs incurred by customers depend on both the duration and frequency or number of interruptions [29], as reflected in the quality-of-service regulations for power distribution [30]. Thus, in order to quantify the interruption costs, $CICE$ and $CICF$ are considered to account for the dependency of customer interruption costs on the duration and frequency of interruptions, respectively. (2) determines the present value factor for customer interruption costs, δ . The total investment and operating costs of the installed switches (all three types) and tie lines are determined in (3). Also, those of the installed switching stations are calculated through the equation set (4)–(8). This set of equations ensures that a switching station is placed at node n if at least one RCS or FCB is installed on the feeder sections or tie lines, for which node n is either the sending or receiving end. In the case this happens for one node, the investment and operation costs of one switching station, IO^{St} , would be added to the total cost of switching stations, formulated in (4). Note that the optimization solver would set the values of $c_n^{St,k}$ to their lower bounds as the objective function is monotonically increasing with respect to them, thereby $c_n^{St,k}$ is set to 0 if no switch of type k is installed at node n . Finally, (9) and (10) respectively calculate the energy-related and frequency-related interruption costs of the customers according to the duration and frequency of their annual power interruptions, denoted by τ

and ν .

$$\min OF = IO^{Sw} + IO^{St} + \delta (CICE + CICF) \quad (1)$$

$$\delta = \frac{1+g}{g-\alpha} \cdot \left(\left(\frac{1+g}{1+\alpha} \right)^T - 1 \right) \quad (2)$$

$$IO^{Sw} = \sum_{l \in L} \sum_{e \in E} \sum_{k \in K} x_{l,e}^k IO^k + \sum_{r \in R} \left(x_r^{Tl} IO_r^{Tl} + \sum_{e \in E} \sum_{k \in K} x_{r,e}^k IO^k \right) \quad (3)$$

$$IO^{St} = \sum_{n \in N} \sum_{k \in \{F, R\}} c_n^{St, k} \quad (4)$$

$$c_n^{St, k} \geq x_{l,es}^k IO^{St, k}; (\forall l \in L, \forall n \in N) | ES_{l,n} = 1, \quad \forall k \in \{F, R\} \quad (5)$$

$$c_n^{St, k} \geq x_{l,er}^k IO^{St, k}; (\forall l \in L, \forall n \in N) | ER_{l,n} = 1, \quad \forall k \in \{F, R\} \quad (6)$$

$$c_n^{St, k} \geq x_{r,es}^k IO^{St, k}; (\forall r \in R, \forall n \in N) | ES_{r,n} = 1, \quad \forall k \in \{F, R\} \quad (7)$$

$$c_n^{St, k} \geq x_{r,er}^k IO^{St, k}; (\forall r \in R, \forall n \in N) | ER_{r,n} = 1, \quad \forall k \in \{F, R\} \quad (8)$$

$$CICE = \sum_{n \in N} \left[\left(\sum_{l \in L} \tau_{l,n} + \sum_{r \in R} \tau_{r,n} \right) D_n \gamma_n^E \right] \quad (9)$$

$$CICF = \sum_{n \in N} \left[\left(\sum_{l \in L} \nu_{l,n} + \sum_{r \in R} \nu_{r,n} \right) D_n \gamma_n^F \right] \quad (10)$$

The proposed optimization model considers some important logical constraints necessary for switch and tie line planning in (11)–(17). Equation (11) ensures that no switch is located at tie line r if the tie line itself is not installed. Equation (12) specifies that if a tie line is built, one of the two ends must be kept normally-open, whereas if not built, both are considered open. Installing an FCB rather than an RCS on a normally-open end would bring no additional benefit for customers' power restoration while it costs more, thus, (13) ensures no FCB would be installed on the normally-open end of an installed tie line. Yet, a switch of another type (an MS or RCS) has to be installed on the normally-open end of an installed tie line, which is considered in (14) and (15). Lastly, (16) and (17) ensure that only one type of switch can be installed at each candidate location. The last two equations are redundant since, for example, installing both an RCS and an MS at the same location is always suboptimal over installing just an RCS. Despite this, including these constraints can enhance the branch-and-cut algorithm's efficiency by pruning the search tree, eliminating suboptimal regions, and potentially accelerating convergence to the global optimum.

$$\sum_{k \in K} x_{r,e}^k \leq x_r^{Tl}; \forall r \in R, \forall e \in E \quad (11)$$

$$\sum_{e \in E} y_{r,e}^{NO} = 2 - x_r^{Tl}; \forall r \in R \quad (12)$$

$$x_{r,e}^F \leq 1 - y_{r,e}^{NO}; \forall r \in R, \forall e \in E \quad (13)$$

$$x_{r,es}^R + x_{r,es}^M \geq y_{r,es}^{NO} - y_{r,er}^{NO}; \forall r \in R \quad (14)$$

$$x_{r,er}^R + x_{r,er}^M \geq y_{r,er}^{NO} - y_{r,es}^{NO}; \forall r \in R \quad (15)$$

$$\sum_{k \in K} x_{l,e}^k \leq 1; \forall l \in L, \forall e \in E \quad (16)$$

$$\sum_{k \in K} x_{r,e}^k \leq 1; \forall r \in R, \forall e \in E \quad (17)$$

B. Reliability Assessment Model

In this subsection, we explain in detail how the annual duration and frequency of customers' power interruptions, i.e., τ and ν , are calculated in the optimization problem.

1) *Calculating ν* : Equations (18)–(24) calculate the annual frequency of power interruptions of customers at node n due to a faulted line section l , which is denoted by $\nu_{l,n}$. Equation (18) specifies that if line section l and load node n are in the same feeder and there is no restoration scheme to isolate load node n from faulted section l , the annual frequency of interruptions, $\nu_{l,n}$, is equal to the failure rate of l , λ_l . However, if there was at least one restoration scheme for load n when feeder section l fails, we first need to calculate the annual frequency of power interruptions at load node n for each possible restoration scheme a that can be used to isolate l . The annual frequency of interruptions for each restoration scheme is denoted by $\phi_{l,n,a}$, which are nonnegative variables as expressed by (19). Therefore, $\nu_{l,n}$ can be easily determined by finding the minimum among the values of $\phi_{l,n,a}$ for all restoration schemes. This is simply because in case of a fault, the load is restored through the best restoration scheme that has the lowest $\phi_{l,n,a}$.

In this regard, if restoration scheme a for load node n is carried out by means of a tie line, the interruption frequency of this scheme will be the same as the failure rate of faulted section l since, in the case of a failure, the customers' power would be interrupted (regardless of installed switches), as the power should be restored through a tie line (all tie lines have a normally-open end that are equipped with either an MS or RCS). This case is formulated in (20). On the contrary, (21) ensures that if no tie line is used in restoration scheme a , the power can be restored without interruptions, subject to the existence of at least one FCB that is able to isolate the fault from the load node instantly. This means if no FCB is installed at the locations that can isolate the fault from the load, $\phi_{l,n,a}$ is equal to λ_l . Note that only FCBs among the available three switch options are able to decrease the frequency of interruptions at load nodes. Also, it is worth mentioning that as the objective function is increasing with respect to $\phi_{l,n,a}$, they would be set to their tightest lower bounds. Lastly, we select the restoration scheme with the lowest interruption frequency, $\phi_{l,n,a}$, as the best restoration scheme, $\nu_{l,n}$, for load node n when section l fails, i.e., $\nu_{l,n} = \min\{\phi_{l,n,a} | \forall a \in A_{l,n}\}$. This is linearly formulated by (22)–(24) using auxiliary binary

variables, $\chi_{l,n,a}$.

$$\nu_{l,n} = \lambda_l; (\forall l \in L, \forall n \in N) | (\xi_{l,n} = 1 \text{ and } A_{l,n} = \emptyset) \quad (18)$$

$$\phi_{l,n,a} \geq 0; \forall l \in L, \forall n \in N \quad (19)$$

$$\phi_{l,n,a} \geq \lambda_l; \forall l \in L, \forall n \in N, \forall a \in A_{l,n} | R_a \neq \emptyset \quad (20)$$

$$\phi_{l,n,a} \geq \lambda_l \left(1 - \sum_{e \in E_a} x_{l,e}^F - \sum_{\bar{l} \in L_a} \sum_{e \in E} x_{\bar{l},e}^F \right); \quad (21)$$

$$\forall l \in L, \forall n \in N, \forall a \in A_{l,n} | R_a = \emptyset \quad (21)$$

$$\nu_{l,n} \leq \phi_{l,n,a}; \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \quad (22)$$

$$\nu_{l,n} \geq \phi_{l,n,a} - M(1 - \chi_{l,n,a}); \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \quad (23)$$

$$\sum_{a \in A_{l,n}} \chi_{l,n,a} = 1; \forall l \in L, \forall n \in N \quad (24)$$

In a similar manner, (26)–(30) calculate the annual frequency of power interruptions of customers at node n due to faulted tie line r , which is denoted by $\nu_{r,n}$. First, the annual frequency of power interruptions at node n for restoration scheme a if tie line r fails, denoted by $\phi_{r,n,a}$, is determined by (25)–(27). Accordingly, (25) expresses the nonnegativity of $\phi_{r,n,a}$. In the case of faulted tie line r and the restoration scheme including another tie line, (26) sets interruption frequency $\phi_{r,n,a}$ to the failure rate of r , λ_r , if the load node is at the feeder connected to the normally-closed end of the faulted tie line. It is trivial that if the load's feeder is connected to the normally-open end, it won't be interrupted. Also, (27) considers that when the power restoration of load node n does not include a tie line when r fails, the load's power will not be interrupted only if either the faulted tie line is connected to the load's feeder through its normally-open end, or there is at least one FCB installed at the locations that can isolate the fault from the node. Otherwise, it will be interrupted, and interruption frequency $\phi_{r,n,a}$ will be the same as λ_r . Finally, (28)–(30) linearly determine the interruption frequency for load node n in case of a fault in tie line r , i.e., $\nu_{r,n}$, which is the lowest value of $\phi_{r,n,a}$ for all fault isolation ways a .

$$\phi_{r,n,a} \geq 0; \forall r \in R, \forall n \in N \quad (25)$$

$$\phi_{r,n,a} \geq \lambda_r \left(1 - \sum_{e \in \bar{E}_a} y_{r,e}^{NO} \right); \quad (26)$$

$$\forall r \in R, \forall n \in N, \forall a \in A_{r,n} | R_a \neq \emptyset \quad (26)$$

$$\phi_{r,n,a} \geq \lambda_r \left(1 - \sum_{e \in \bar{E}_a} y_{r,e}^{NO} - \sum_{e \in E_a} x_{r,e}^F - \sum_{\bar{l} \in L_a} \sum_{e \in E} x_{\bar{l},e}^F \right); \quad (27)$$

$$\forall r \in R, \forall n \in N, \forall a \in A_{r,n} | R_a = \emptyset \quad (27)$$

$$\nu_{r,n} \leq \phi_{r,n,a}; \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \quad (28)$$

$$\nu_{r,n} \geq \phi_{r,n,a} - M(1 - \chi_{r,n,a}); \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \quad (29)$$

$$\sum_{a \in A_{r,n}} \chi_{r,n,a} = 1; \forall r \in R, \forall n \in N \quad (30)$$

2) *Calculating τ* : Equations (31)–(42) find the annual duration of power interruptions of customers at node n due to the failure of feeder section l , i.e., denoted by $\tau_{l,n}$. Equation (31) determines the annual duration of power interruptions, $\tau_{l,n}$ whenever feeder section l and load node n are in the same feeder and there is no restoration scheme to isolate load node n from faulted section l . Nonetheless, similar to calculating $\nu_{l,n}$, if there exists at least one restoration scheme for load node n to isolate it from faulted section l , we should first determine the annual interruption duration of load node n due to faulted section l for all possible restoration schemes, denoted by $\theta_{l,n,a}$. Then, $\tau_{l,n}$ can be readily determined by finding the minimum among the values of $\theta_{l,n,a}$ for all restoration schemes as it shows the fastest restoration scenario. It is worth remembering that as the objective function is increasing with respect to $\theta_{l,n,a}$, they would be set to their tightest lower bounds, determined through the following set of equations. Accordingly, (32) implies the nonnegativity of $\theta_{l,n,a}$. Expression (33) ensures that if the restoration of power is done through a tie line, the minimum restoration time is equal to the switching time of an RCS since, even in the best case, the normally-open side of the tie line is equipped with an RCS, through which the power should pass to energize the load. Also, (34) specifies that if no tie line is required, the restoration time can be zero when at least an FCB is installed at the locations that can isolate the fault from the load. If not, the restoration would take at least as long as the RCS switching time. Equation (35) sets the minimum restoration time to the MS switching time when the restoration should be done through a tie line but there is no RCS installed at the locations that can isolate the fault from the load. Note that in such cases that the load nodes are “downstream” of a faulted section (i.e., the faulted section does not allow reenergizing the load from the main supply and requires a tie line for restoration), the FCB(s) installed at locations that can isolate the fault from the load will not operate automatically as the fault current does not pass through such FCBs. Yet, they can be used manually, thus, under such circumstances, they are modeled similar to MSs in this optimization model. Equation (36) specifies that if the restoration scheme does not include a tie line but there is no FCB or RCS to isolate the fault from the load, the minimum restoration time is equal to the MS switching time. Through (37), we ensure that the restoration time cannot be lower than the MS switching time, if the power restoration of load should go through a tie line while the switch installed on its normally-open end is not an RCS. For all restoration schemes, if no switch is installed at the locations that can be opened to isolate the fault from the load, the minimum restoration time is equal to the repair time of faulted section l , which is formulated in (38). Also, the minimum restoration time will be equal to the repair time of faulted section if the tie line that can be used to reenergize the load is not installed, as formulated in (39). Note that if there is an installed tie line at r , (14) and (15) enforce the installation of at least an MS on the normally-open end. Finally, (40)–(42) calculate the interruption duration for customers at load node n

when section l fails, $\tau_{l,n}$, by finding the minimum restoration duration among all possible scenarios, $\theta_{l,n,a}$.

$$\tau_{l,n} = \lambda_l \pi_l; (\forall l \in L, \forall n \in N) \mid (\xi_{l,n}=1 \text{ and } A_{l,n}=\emptyset) \quad (31)$$

$$\theta_{l,n,a} \geq 0; \forall l \in L, \forall n \in N \quad (32)$$

$$\theta_{l,n,a} \geq \lambda_l \pi^R; \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \mid R_a \neq \emptyset \quad (33)$$

$$\theta_{l,n,a} \geq \lambda_l \pi^R \left(1 - \sum_{e \in E_a} x_{l,e}^F - \sum_{\bar{l} \in L_a} \sum_{e \in E} x_{\bar{l},e}^F \right); \quad \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \mid R_a = \emptyset \quad (34)$$

$$\theta_{l,n,a} \geq \lambda_l \pi^M \left(1 - \sum_{e \in E_a} x_{l,e}^R - \sum_{\bar{l} \in L_a} \sum_{e \in E} x_{\bar{l},e}^R \right); \quad \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \mid R_a \neq \emptyset \quad (35)$$

$$\theta_{l,n,a} \geq \lambda_l \pi^M \left(1 - \sum_{e \in E_a} \sum_{k \in \{F,R\}} x_{l,e}^k - \sum_{\bar{l} \in L_a} \sum_{e \in E} \sum_{k \in \{F,R\}} x_{\bar{l},e}^k \right); \quad \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \mid R_a = \emptyset \quad (36)$$

$$\theta_{l,n,a} \geq \lambda_l \pi^M \left(1 - \sum_{r \in R_a} \sum_{e \in E} x_{r,e}^R y_{r,e}^{NO} \right); \quad \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \mid R_a \neq \emptyset \quad (37)$$

$$\theta_{l,n,a} \geq \lambda_l \pi_l \left(1 - \sum_{e \in E_a} \sum_{k \in K} x_{l,e}^k - \sum_{\bar{l} \in L_a} \sum_{e \in E} \sum_{k \in K} x_{\bar{l},e}^k \right); \quad \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \quad (38)$$

$$\theta_{l,n,a} \geq \lambda_l \pi_l \left(1 - \sum_{r \in R_a} x_r^{TL} \right); \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \mid R_a \neq \emptyset \quad (39)$$

$$\tau_{l,n} \leq \theta_{l,n,a}; \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \quad (40)$$

$$\tau_{l,n} \geq \theta_{l,n,a} - M (1 - \chi_{l,n,a}^T); \forall l \in L, \forall n \in N, \forall a \in A_{l,n} \quad (41)$$

$$\sum_{a \in A_{l,n}} \chi_{l,n,a}^T = 1; \forall l \in L, \forall n \in N \quad (42)$$

Similar to the previous set of equations, (43)–(52) determine the annual duration of power interruptions of customers at node n due to faulted tie line r , which is denoted by $\tau_{r,n}$. Thus, as explained before, we should calculate the annual interruption duration of load node n due to faulty tie line r for all possible restoration schemes, shown by $\theta_{r,n,a}$ in the following equations. First, note that all the following equations are for the scenario that the load node n is at a feeder that is connected to the normally-closed end of faulted tie line r . It goes without saying that if the load's feeder is connected to the normally-open end of the tie line, it will not be interrupted. In this context, (43) specifies that if tie line r fails and the restoration scheme of load node n is through another tie line, the minimum restoration time

of load is the RCS switching time. Also, when the restoration does not include a tie line, (44) sets the restoration time of load node n when r fails to the RCS switching time if there is no FCB installed at the locations that can isolate the faulted tie line from the node. Equation (45) sets the minimum restoration time to the MS switching time when reenergizing the load includes a tie line while there is no RCS installed between the faulted tie line and the load. In this case, FCBs that can isolate the fault are operated manually, as was explained for (35). For the restoration scenarios that do not include any tie line, (46) specifies the minimum restoration time equal to the MS switching time if there is no RCS or FCB that can isolate the faulted tie line from the load node. Equation (47) ensures that if none of the switches installed on the normally-open ends of the tie lines required for load restoration is an RCS, the restoration time cannot be lower than the MS switching time. In all restoration schemes, if no switch is installed at the locations that can be used to isolate the fault from the load, (48) sets the minimum restoration time to the repair time of faulted tie line r , i.e., π_r . Equation (49) ensures that the restoration time is set to π_r when the restoration can be done through a candidate tie line but it is not installed. Lastly, (50)–(52) determine the interruption duration for customers at load node n when tie line r fails, $\tau_{r,n}$, by finding the minimum value of $\theta_{r,n,a}$ for all possible restoration schemes.

$$\theta_{r,n,a} \geq \lambda_r \pi^R \left(1 - \sum_{e \in \bar{E}_a} y_{r,e}^{NO} \right); \quad \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \mid R_a \neq \emptyset \quad (43)$$

$$\theta_{r,n,a} \geq \lambda_r \pi^R \left(1 - \sum_{e \in \bar{E}_a} y_{r,e}^{NO} - \sum_{e \in \bar{E}_a} x_{r,e}^F - \sum_{\bar{l} \in L_a} \sum_{e \in E} x_{\bar{l},e}^F \right); \quad \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \mid R_a = \emptyset \quad (44)$$

$$\theta_{r,n,a} \geq \lambda_r \pi^M \left(1 - \sum_{e \in \bar{E}_a} y_{r,e}^{NO} - \sum_{e \in \bar{E}_a} x_{r,e}^R - \sum_{\bar{l} \in L_a} \sum_{e \in E} x_{\bar{l},e}^R \right); \quad \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \mid R_a \neq \emptyset \quad (45)$$

$$\theta_{r,n,a} \geq \lambda_r \pi^M \left(1 - \sum_{e \in \bar{E}_a} y_{r,e}^{NO} - \sum_{k \in \{F,R\}} \left(\sum_{e \in \bar{E}_a} x_{r,e}^k - \sum_{\bar{l} \in L_a} \sum_{e \in E} x_{\bar{l},e}^k \right) \right); \quad \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \mid R_a = \emptyset \quad (46)$$

$$\theta_{r,n,a} \geq \lambda_r \pi^M \left(1 - \sum_{e \in \bar{E}_a} y_{r,e}^{NO} - \sum_{\bar{r} \in R_a} \sum_{e \in E} x_{\bar{r},e}^R y_{\bar{r},e}^{NO} \right); \quad \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \mid R_a \neq \emptyset \quad (47)$$

$$\theta_{r,n,a} \geq \lambda_r \pi_r \left(1 - \sum_{e \in \bar{E}_a} y_{r,e}^{NO} - \sum_{k \in K} \left(\sum_{e \in \bar{E}_a} x_{r,e}^k - \sum_{\bar{l} \in L_a} \sum_{e \in E} x_{\bar{l},e}^k \right) \right); \quad \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \quad (48)$$

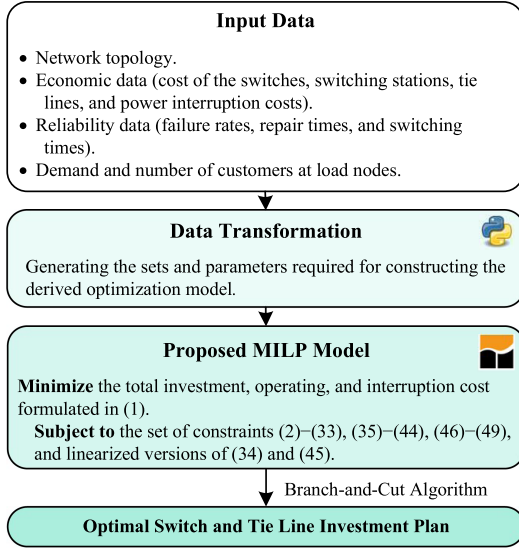


Fig. 2. Flowchart of the solution process.

$$\theta_{r,n,a} \geq \lambda_r \pi_r \left(1 - \sum_{e \in \bar{E}_a} y_{r,e}^{NO} - \sum_{\bar{r} \in R_a} x_{\bar{r}}^{Tl} \right);$$

$$\forall r \in R, \forall n \in N, \forall a \in A_{r,n} | R_a \neq \emptyset \quad (49)$$

$$\tau_{r,n,a} \leq \theta_{r,n,a}; \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \quad (50)$$

$$\tau_{r,n,a} \geq \theta_{r,n,a} - M(1 - \chi_{r,n,a}^\tau); \forall r \in R, \forall n \in N, \forall a \in A_{r,n} \quad (51)$$

$$\sum_{a \in A_{r,n}} \chi_{r,n,a}^\tau = 1; \forall r \in R, \forall n \in N \quad (52)$$

C. Solution Process

To use a commercial MILP solver for solving the optimization problem, we should avoid multiplications of binary variables, i.e., $x_{r,e}^R y_{r,e}^{NO}$ and $x_{\bar{r},e}^R y_{\bar{r},e}^{NO}$ respectively in (37) and (47), which can be done by introducing new auxiliary variables representing the multiplication terms. For instance, multiplication of two binary variables x and y is done by considering an auxiliary variable z and the following constraints:

$$\begin{aligned} z &\leq x \\ z &\leq y \\ z &\geq x + y - 1 \end{aligned}$$

The flowchart in Fig. 2 describes how the proposed MILP model for the optimal switch and tie line planning is utilized. As per this flowchart, the process includes a preprocessing stage implemented in the Python programming language that gets the input data and delivers the required sets for the MILP optimization, i.e., $\xi_{l,n}$, $A_{l,n}$, E_a , R_a , and L_a . Finally, the MILP model as described in the previous two sections is solved using an off-the-shelf MILP solver. In this paper, the IBM CPLEX solver under the GAMS modeling tool is used.

III. HEURISTIC APPROACH FOR CLOSE TO OPTIMAL SWITCH PLANNING

The underlying logic for the crude but fast heuristic switch optimization technique [28], which is available as a module in a Network Information System used by many Nordic distribution utilities, is to locate MSs at all candidate locations, and then to make benefit-to-cost decisions about each switch one at a time. First the switch in each main feeder section that has the most net saving (cost benefit to interruption costs) is removed if optimal rather than full switching is stipulated by the user. Then the remaining switches are checked for upgrade to remote and then further upgrade to FCBs.

As presented in Algorithm 1, switches are first placed at each end of every feeder section, including tie lines, except the sending end of each main feeder, which is by default equipped with a main circuit breaker. Utilizing parallel processing for each feeder, the algorithm first decides where MSs should be kept by removing the MS with the maximum negative cost-to-benefit at each iteration, and then considers which of the remaining MSs should be upgraded to RCSs. Then, the RCSs are upgraded to FCBs in order of benefit to cost. If full switching is not required, a final sweep is made to remove any MSs where the costs outweigh the benefits, to arrive at a close-to-optimal switch allocation solution. The heuristic algorithm assesses the impact of faults in every relevant network component in terms of which customers suffer an interruption, and for how long (repair and/or manual or remote switching time). Computational efficiency is enhanced by using delta-cost functions for every change in switching topology, rather than computing full interruption costs for each topology.

Note that finding the best option for cost saving at each step is not trivial, as it involves considering every possible fault in each feeder, the nodes that are affected by the fault, and for what length of time they are deprived electricity. However the heuristic constraints and parallel processing give very fast computation times, which makes the method suitable for embedding in a planning algorithm, whereas the MILP algorithm proposed in this paper is highly suited for both post-processing in a planning algorithm and ascertaining how close-to-optimum the fast heuristic algorithm actually is.

IV. NUMERICAL RESULTS

In this section, to numerically demonstrate the applicability of our MILP model and to benchmark the performance and accuracy of the heuristic approach against the MILP model, we report and discuss the simulation results of implementing it on a few test distribution networks under different assumptions. The MILP model was implemented in GAMS 27.3.0, and IBM CPLEX 12.9 is utilized as the solver with an optimality gap of 0% as the stopping criterion. The simulations have been done on a Dell Precision 3650 Tower PC with a 6 Core 2.80GHz Intel Core i5-11600 processor and 32GB of RAM. Also, the input data used for our simulations, including the test distribution network details, are available at [31].

Algorithm 1: Heuristic Algorithm for the Switch Placement Problem.

```

for each feeder do
  repeat
    for each remaining MS in the feeder do
      if MS investment > interruption cost saving then
        Log the net cost saving of removing the MS
      end
    end
    Remove the MS that provides the greatest saving
  until no MS satisfies the removal condition
  repeat
    for each remaining MS in the feeder do
      if cost for manual-to-remote switch upgrade <
        interruption cost saving then
        Log the net cost saving of upgrading the MS to
        RCS
      end
    end
    Upgrade the switch that provides the greatest saving
  until no MS satisfies the upgrade condition
  repeat
    for each remaining RCS in the feeder do
      if cost for RCS to FCB upgrade < interruption cost
        saving then
        Log the net cost saving of upgrading the RCS to
        FCB
      end
    end
    Upgrade the switch that provides the greatest saving
  until no RCS satisfies the upgrade condition
  repeat
    for each remaining MS in the feeder do
      if cost for removing MS < interruption cost saving
        then
        Log the net cost saving of removing the MS
      end
    end
    Remove the MS that yields the greatest saving
  until no MS satisfies the removal condition
end

```

A. Testing the MILP Model Applicability on a 54-Node System

To validate the applicability of the proposed MILP model and demonstrate the significance of its practical features, we implemented it on a 54-node test distribution network, consisting of 50 load nodes, 4 substations, and 11 candidate tie line locations. To evaluate the significance of the model's features, we considered three different cases in our simulations:

- Case I: A modified version of the proposed model in which installing new tie lines is eliminated.
- Case II: Proposed model.
- Case III: A modified version of the proposed model where tie line failures are disregarded.

TABLE II
NUMERICAL RESULTS – 54-NODE TEST NETWORK

	Case I	Case II	Case III
Objective function (k\$)	3,240.414	1,591.203	1,536.743
IO^{Sw} (k\$)	236.200	955.570	914.070
IO^{St} (k\$)	8.000	14.400	12.000
$\delta(CICE + CICF)$ (k\$)	2,996.214	621.233	610.673
SAIFI (int./cust./year)	0.147	0.157	0.147
SAIDI (hours/cust./year)	0.741	0.141	0.140
EENS (MWh/year)	10.124	1.921	1.900
No. of installed MSs	14	36	33
No. of installed RCSs	13	23	21
No. of installed FCBs	0	0	0
No. of installed tie lines	-	5	5

First, the solver could find the global optimal solutions in just 0.76, 2.20, and 1.42 seconds respectively for Cases I-III, demonstrating the high efficiency of the MILP model. Table II presents the results of these simulations. Comparing the results of Cases I and II, it is evident that not installing new tie lines have led to significantly worse SAIDI and EENS, and thus a much higher objective function, despite the fewer installed switches and tie lines in Case I. Consistent with our expectations, this result demonstrates that considering the installation of tie lines in the switch optimization problem is vital to achieve the best possible system reliability.

Furthermore, by comparing the results for Case III with those of Case II, we can understand that eliminating the impact of tie line failures will result in an overoptimistic solution, where the interruption costs are underestimated. Consequently, less investments are made on SSs in Case III, leading to a less effective switch arrangement compared to Case II. This is numerically validated by calculating the actual objective function and reliability metrics (i.e., when tie line failures are considered) for the switch arrangement solution obtained in Case III: Objective function = 1,804.948k\$, SAIFI = 0.157, SAIDI = 0.190, EENS = 2.806, which are worse than those in Case II. It is worth mentioning that the optimal solutions in all three cases do not include installing any FCBs since the test network consists of several short feeders. As a result, it is not profitable to install FCBs on any feeder.

B. Comparison With the Heuristic Model

Although the proposed MILP model concurrently optimizes switch and tie line planning, the heuristic model, described in Section III, only performs the optimal SS and FCB allocation. This is because tie line planning is done by a different module in the planning algorithm of which the heuristic approach is a part. Therefore, for the sake of making the MILP and heuristic models comparable, the arrangement of tie lines is assumed to be known in advance in the simulations of this subsection; nonetheless, the investment in switches of the tie lines is optimized by the models.

1) *37-Node System:* To compare the switch plan solutions obtained by the heuristic algorithm and the proposed MILP model, we implemented both on a 37-node distribution network. The solutions obtained from the two approaches are illustrated in Figs. 3 and 4, respectively. Also, the numerical results are presented in Table III. The solution time for the MILP model

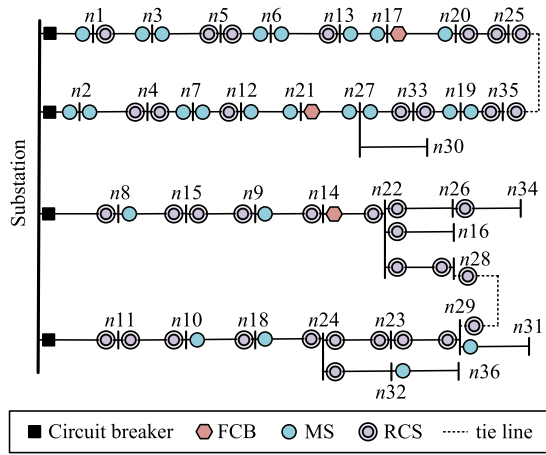


Fig. 3. Resulting switch plan for the 37-node network – heuristic model.

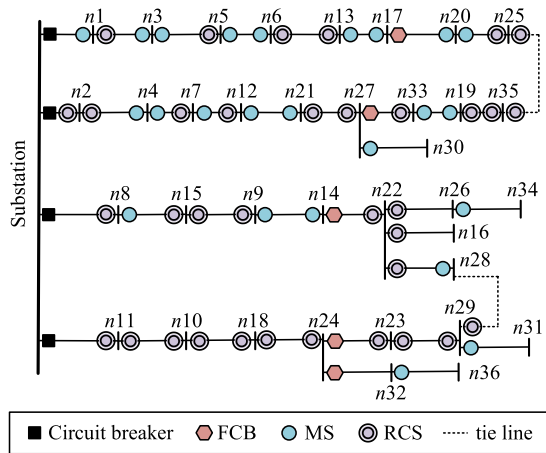


Fig. 4. Global optimal solution obtained using the proposed MILP model for switch planning of the 37-node network.

TABLE III
NUMERICAL RESULTS – 37-NODE TEST NETWORK

	MILP	Heuristic
Objective function (k\$)	1,736.431	1,862.342
IO^{Sw} (k\$)	680.900	652.700
IO^{St} (k\$)	22.400	20.600
$\delta(CICE + CICF)$ (k\$)	1,033.131	1,189.042
SAIFI (int./cust./year)	0.261	0.289
SAIDI (hours/cust./year)	0.140	0.151
EENS (MWh/year)	5.945	6.853
No. of installed MSs	24	24
No. of installed RCSs	35	37
No. of installed FCBs	5	3

was 0.45 seconds, and the heuristic method reached a solution in a few milliseconds. As our proposed model is an instance of MILP, the solver ensures achieving the global optimal solution of the problem considering that we set the optimality tolerance to 0. However, the solution reached by the heuristic algorithm is not global optimal. What is worse is that the distance between its solution and the global optimum depends on the problem; sometimes they can be close and sometimes they might not. In

TABLE IV
OUTCOMES FOR THE MILP AND HEURISTIC APPROACHES

Test network Approach	186-node		230-node	
	MILP	Heuristic	MILP	Heuristic
Objective function (k\$)	2,979.6	3,110.6	3,983.4	4,038.9
IO^{Sw} (k\$)	1,057.4	1,021.7	2,349.4	2,378.0
IO^{St} (k\$)	23.6	22.0	59.6	59.4
$\delta(CICE + CICF)$ (k\$)	1,898.7	2,066.9	1,574.4	1,601.5
SAIFI (int./cust./year)	0.624	0.834	0.046	0.048
SAIDI (hours/cust./year)	0.643	0.688	0.031	0.031
EENS (MWh/year)	11.836	122.670	4.646	4.714
Simulation time (s)	61.170	0.145	5.780	0.030

TABLE V
NUMBER OF INSTALLED SWITCHES AND TIE LINES

Test network Approach	186-node		230-node	
	MILP	Heuristic	MILP	Heuristic
No. of MSs	83	83	236	243
No. of RCSs	32	40	74	75
No. of FCBs	7	2	2	1

this case, the objective function got by the heuristic is about 7% worse than the global optimal solution. The difference is mainly due to the fact that all reliability indices (SAIDI, SAIFI, and EENS) of the heuristic solution are worse than those of the MILP. Consequently, despite spending slightly less on installing switches, the obtained solution by the heuristic approach is worse than that of the MILP model. As per Figs. 3 and 4, not only the numbers of installed RCSs and FCBs are different for the two solutions, but also there are differences between their SS arrangements. This shows the importance of finding the best possible switch arrangement, obtained by the MILP model, in achieving better reliability indices for the distribution network.

2) *186- and 230-Node Networks:* While the benefit of the proposed MILP technique in obtaining better solutions was shown, a big drawback of MILP models in general is their computational burden. Thus, many in industry prefer heuristic methods as, despite their lower quality solutions, they are efficient and can find a solution in an acceptable amount of time. Here, we will show the scalability and high efficiency of the proposed MILP model, which makes it practical for real-world networks. In this regard, we applied both the heuristic algorithm and the MILP model to two relatively large distribution networks, a 186-node rural and a 230-node urban network. The results obtained by both methods are presented in Tables IV and V. First, as could be expected, the MILP once again finds better solutions for the problem, compared to the heuristic. However, the solution that the heuristic finds for the 230-node network is more accurate (i.e., closer to the global optimum) compared to the one obtained for the 186-node network. While both solutions are relatively accurate (less than 5% worse than the global optimum), the results vividly show that the solution quality changes in different cases. Despite this, we can see in the table, that the heuristic algorithm could find a solution in less than a second, while the MILP took about 6 and 61 seconds to find the solutions of 230-node and 186-node networks, respectively. While this clearly indicates the lower computational burden of the heuristic, the MILP model is still considered highly efficient

TABLE VI
NUMERICAL RESULTS – 1000-NODE TEST NETWORK

	MILP	Heuristic
Objective function (k\$)	18,449.418	18,888.469
IO^{Sw} (k\$)	7,679.300	7,895.100
IO^{St} (k\$)	207.200	218.400
$\delta(CICE + CICF)$ (k\$)	10,562.918	10,672.169
SAIFI (int./cust./year)	0.085	0.085
SAIDI (hours/cust./year)	0.127	0.129
EENS (MWh/year)	33.587	33.963
No. of installed MSs	721	739
No. of installed RCSs	274	292
No. of installed FCBs	4	4

as well, taking just about one minute to find the guaranteed global optimal solution of this problem. This means that while solving the problem for a large network may take much longer (e.g., in the order of a few hours), the problem would still be tractable and can be solved in a finite amount of time, which is actually the issue that matters to distribution network planners. This high efficiency, together with the practical features discussed before, makes the proposed MILP model applicable for switch planning purposes in real-world distribution networks.

3) *1000-Node Network*: In order to investigate the scalability of the proposed MILP model, it is applied to a distribution system with 1000 nodes. With the optimality gap set to 0, the MILP model is solved in 50.88 seconds, further demonstrating the scalability and computational performance of the proposed modeling framework. The results obtained from both the MILP model and the heuristic approach are presented in Table VI. As per this table, the total cost of the solution obtained by the heuristic approach is 2.38% higher than that of the global optimal solution found by the proposed MILP model. This, in turn, reveals the acceptable solution quality of the heuristic technique, which only required one-tenth of a second to solve.

V. CONCLUSION

In this paper, we have proposed a novel MILP model for the allocation of SSs, FCBs, and tie lines in networks with complex topologies. In this model, we avoid the simplifying assumptions made by many state-of-the-art studies, providing the most accurate MILP model for switch allocation in distribution networks. Due to the high efficiency of the proposed model, the optimal switch placement problem can be solved in a finite amount of time, while guaranteeing convergence to the global optimum. These features make the MILP model suitable for practical applications and comparable with the existing heuristic methods used by industry to solve this problem for real-world distribution networks. For comparison, we presented an efficient heuristic algorithm that is currently used by distribution companies for planning purposes. Implementing the proposed MILP model on 5 different networks, we demonstrated its applicability in realistic distribution networks and its advantage over the heuristic method by comparing their solutions. Additionally, we showed how some of the common assumptions made by other state-of-the-art MILP models would lead to non-optimal solutions. We

believe, as the proposed model in this paper considers the most important features in a real-world switch planning problem and can be easily formulated and solved by off-the-shelf solvers, distribution utilities will be able to use it for planning purposes, achieving better solutions with less overall costs. The proposed MILP model could also be used for post-processing in a planning tool, if it is still found to be too computationally burdensome for full integration in a planning algorithm.

REFERENCES

- [1] R. J. Millar, S. Kazemi, M. Lehtonen, and E. Saarijärvi, "Impact of MV connected microgrids on MV distribution planning," *IEEE Trans. Smart Grid*, vol. 3, no. 4, pp. 2100–2108, Dec. 2012.
- [2] R. Billinton and S. Jonnavithula, "Optimal switching device placement in radial distribution systems," *IEEE Trans. Power Del.*, vol. 11, no. 3, pp. 1646–1651, Jul. 1996.
- [3] G. Levitin, S. Mazal-Tov, and D. Elmakis, "Optimal sectionalizer allocation in electric distribution systems by genetic algorithm," *Elect. Power Syst. Res.*, vol. 31, no. 2, pp. 97–102, Nov. 1994.
- [4] J.-H. Teng and Y.-H. Liu, "A novel ACS-based optimum switch relocation method," *IEEE Trans. Power Syst.*, vol. 18, no. 1, pp. 113–120, Feb. 2003.
- [5] C.-S. Chen, C.-H. Lin, H.-J. Chuang, C.-S. Li, M.-Y. Huang, and C.-W. Huang, "Optimal placement of line switches for distribution automation systems using immune algorithm," *IEEE Trans. Power Syst.*, vol. 21, no. 3, pp. 1209–1217, Aug. 2006.
- [6] Y. Mao and K. N. Miu, "Switch placement to improve system reliability for radial distribution systems with distributed generation," *IEEE Trans. Power Syst.*, vol. 18, no. 4, pp. 1346–1352, Nov. 2003.
- [7] A. Abiri-Jahromi, M. Fotuhi-Firuzabad, M. Parvania, and M. Mosleh, "Optimized sectionalizing switch placement strategy in distribution systems," *IEEE Trans. Power Del.*, vol. 27, no. 1, pp. 362–370, Jan. 2012.
- [8] L. S. de Assis et al., "Switch allocation problems in power distribution systems," *IEEE Trans. Power Syst.*, vol. 30, no. 1, pp. 246–253, Jan. 2015.
- [9] T. Zhang, C. Wang, F. Luo, P. Li, and L. Yao, "Optimal design of the sectional switch and tie line for the distribution network based on the fault incidence matrix," *IEEE Trans. Power Syst.*, vol. 34, no. 6, pp. 4869–4879, Nov. 2019.
- [10] O. K. Siirto, A. Safdarian, M. Lehtonen, and M. Fotuhi-Firuzabad, "Optimal distribution network automation considering earth fault events," *IEEE Trans. Smart Grid*, vol. 6, no. 2, pp. 1010–1018, Mar. 2015.
- [11] A. Heidari, V. G. Agelidis, and M. Kia, "Considerations of sectionalizing switches in distribution networks with distributed generation," *IEEE Trans. Power Deliv.*, vol. 30, no. 3, pp. 1401–1409, Jun. 2015.
- [12] M. Izadi and A. Safdarian, "A MIP model for risk constrained switch placement in distribution networks," *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 4543–4553, Jul. 2019.
- [13] M. Farajollahi, M. Fotuhi-Firuzabad, and A. Safdarian, "Optimal placement of sectionalizing switch considering switch malfunction probability," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 403–413, Jan. 2019.
- [14] M. Farajollahi, M. Fotuhi-Firuzabad, and A. Safdarian, "Simultaneous placement of fault indicator and sectionalizing switch in distribution networks," *IEEE Trans. Smart Grid*, vol. 10, no. 2, pp. 2278–2287, Mar. 2019.
- [15] M. Jooshaki, S. Karimi-Arpanahi, M. Lehtonen, R. J. Millar, and M. Fotuhi-Firuzabad, "Electricity distribution system switch optimization under incentive reliability scheme," *IEEE Access*, vol. 8, pp. 93455–93463, 2020.
- [16] M. Jooshaki, S. Karimi-Arpanahi, M. Lehtonen, R. J. Millar, and M. Fotuhi-Firuzabad, "Reliability-oriented electricity distribution system switch and tie line optimization," *IEEE Access*, vol. 8, pp. 130967–130978, 2020.
- [17] A. Shahbazian, A. Fereidunian, and S. D. Manshadi, "Optimal switch placement in distribution systems: A high-accuracy MILP formulation," *IEEE Trans. Smart Grid*, vol. 11, no. 6, pp. 5009–5018, Nov. 2020.
- [18] Z. Wang, D. Lin, T. Yu, G. Zeng, and Z. Pan, "Explicit reliability incorporated switches planning of smart distribution system with high flexibility," *Int. J. Elect. Power Energy Syst.*, vol. 145, 2023, Art. no. 108629.
- [19] C. Wang, K. Pang, M. Shahidehpour, F. Wen, H. Ren, and Z. Liu, "Flexible joint planning of sectionalizing switches and tie lines among distribution feeders," *IEEE Trans. Power Syst.*, vol. 37, no. 2, pp. 1577–1590, Mar. 2022.

- [20] M. Jooshaki, S. Karimi-Arpanahi, M. Lehtonen, R. J. Millar, and M. Fotuhi-Firuzabad, "An MILP model for optimal placement of sectionalizing switches and tie lines in distribution networks with complex topologies," *IEEE Trans. Smart Grid*, vol. 12, no. 6, pp. 4740–4751, Nov. 2021.
- [21] M. Jooshaki et al., "An enhanced MILP model for multistage reliability-constrained distribution network expansion planning," *IEEE Trans. Power Syst.*, vol. 37, no. 1, pp. 118–131, Jan. 2022.
- [22] A. C. Gomes, R. P. S. Leão, B. de Athayde Prata, F. L. Tofoli, R. F. Sampaio, and G. C. Barroso, "Optimal placement of manual and remote-controlled switches based on the Pareto front," *Int. J. Elect. Power Energy Syst.*, vol. 147, May 2023, Art. no. 108894.
- [23] M. Izadi, M. Farajollahi, and A. Safdarian, "Optimal deployment of remote-controlled switches in distribution networks considering laterals," *IET Gener. Transm. Distrib.*, vol. 13, no. 15, pp. 3264–3271, Aug. 2019.
- [24] M. Jooshaki, S. Karimi-Arpanahi, J. Millar, M. Lehtonen, and M. Fotuhi-Firuzabad, "On the MILP modeling of remote-controlled switch and field circuit breaker malfunctions in distribution system switch placement," *IEEE Access*, vol. 11, pp. 40905–40915, 2023.
- [25] G. Muñoz-Delgado, J. Contreras, and J. M. Arroyo, "Multistage generation and network expansion planning in distribution systems considering uncertainty and reliability," *IEEE Trans. Power Syst.*, vol. 31, no. 5, pp. 3715–3728, Sep. 2016.
- [26] G. Muñoz-Delgado, J. Contreras, and J. M. Arroyo, "Reliability assessment for distribution optimization models: A non-simulation-based linear programming approach," *IEEE Trans. Smart Grid*, vol. 9, no. 4, pp. 3048–3059, Jul. 2018.
- [27] G. Muñoz-Delgado, J. Contreras, and J. M. Arroyo, "Distribution network expansion planning with an explicit formulation for reliability assessment," *IEEE Trans. Power Syst.*, vol. 33, no. 3, pp. 2583–2596, May 2018.
- [28] R. J. Millar, M. Lehtonen, and E. Saarijärvi, "Switch and reserve connection placement in a distribution network planning algorithm," in *Proc. 2012 IEEE Int. Conf. Power Syst. Technol.*, Auckland, New Zealand, Oct. 2012, pp. 1–6.
- [29] R. Billinton and R. N. Allan, *Reliability Evaluation of Power Systems*. New York, NY, USA: Plenum Press, 1984.
- [30] "7th CEER-ECRB benchmarking report on the quality of electricity and gas supply," Council of Eur. Energy Regulators (CEER), Tech. Rep. C22-EQS-103-03, 2022. Accessed: Feb. 2024. [Online]. Available: <https://www.ceer.eu/documents/104400/7324389/7thBenchmarkingReport/15277cb7-3ffe-8498-99bb-6f083e3ceecb>
- [31] M. Jooshaki and J. Millar, "Test distribution systems for optimal switch placement," *Mendeley Data*, V2, 2024. Accessed: Sep. 2024. [Online]. Available: <https://data.mendeley.com/datasets/w5kkt6v88/2>